Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 41 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 41 Moment of Inertia Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 41 Moment of Inertia Numerical

1 CLASS 41 Moment of Inertia Numerical

Study focus: CLASS 41 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the question A slash 42.

Key details

- Calculate the moment of inertia of the shaded area about the x axis.
- We have been asked to find moment of inertia of only this shaded area about the x-axis.
- How to find inertia of this shaded area about the x-axis?
- What we can do here is we can find moment of inertia of this triangle about the x-axis, then in that we can detect the moment of inertia of this semicircle about the same x-axis.
- Then we are going to get moment of inertia of this remaining shaded area.
- About x axis.
- In this, we have to deduct minus moment of inertia of a semicircle.
- Of a semi circle about the same x axis.
- Next, what is moment of inertia effect triangle about x axis?
- Moment of inertia of a triangle about x axis, we know from basic formulas.
- That is equal to, I am going to call that as Ix1.
- That is moment of inertia of a triangle about x axis.
- That is equals to bh cube by.
- This is from the basic formulas we know.
- Here, what is B?
- B is the base of the triangle that is hundred.
- What is H?
- H is height of a triangle that is 90.
- Use the calculator, then find this value.
- That is six point zero seven five into ten per six.

- mm to the power of 4.
- This is the moment of inertia of a triangle about its base, that is about x-axis.
- Next we need to find the inertia of a semicircle about the same x-axis that I call it as Ix2.
- This will be equal to Since the X axis is passing through the.
- Is nothing but the diameter axis.
- We have a direct formula.
- Moment of inertia of a semicircle about its diameter axis is pi r to the power of 4 by 8.
- H r is 30.
- to the power of four by eight.
- Just the calculator find this value.
- This is zero point three one eight into ten power six mm to the power of four.
- This is the moment of inertia of a triangle about x axis minus moment of inertia of this semicircle about the x axis is 0.318 into 10 po 6.
- Once we solve this, we are going to get as 5.756 into 10 bar 6 mm to the power of 4.
- This is the moment of inertia of the shaded area about the x axis.
- This is the final answer.
- X question A slash 44.
- The variable h designates the arbitrary vertical location of the center of a circular cutout within the semicircular area.
- Determine the area moment of inertia about the x-axis for A H is equals to 0 and when H is equals to R by 2.
- In the first case, before that, look at this, there is a semicircle of radius capital R.
- And there is this circular cutout.
- When the position of the center of this circular cutout is h is equal to 0.
- That is when the center of the circle is coinciding with the origin O, in that case, what is the moment of inertia of the.

Example or application from the lecture

- Let's get into the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

Shaded area, nothing but what is the moment of inertia of the remaining area we have to calculate.

Key details

- Then in the second case, when H that is the position of the center of this circular cutout is R by 2, what is the moment of inertia of this?
- Remaining area about the x axis we have to calculate.
- when h is equal to 0.
- When h is equals to 0, how will be the figure?
- Since h is equal to 0, the half of the circle is going to be here.
- And the remaining half it's going to be outside this semicircle.

- The part which is outside the semicircle that doesn't matter to us because we are considering moment of inertia of the remaining area within this semicircle.
- The radius of this semicircle is capital R.
- Then the radius of this circle is R by 4.
- Then we are considering only this shaded.
- You need to find moment of inertia of only this shaded area about the x-axis.
- That will be equal to moment of inertia of this bigger semicircle minus moment of inertia of this smaller semicircle about the x-axis.
- Ix is nothing but moment of inertia of the shaded area about x-axis is equal to moment of inertia of a semicircle.
- About X axis.
- This is because the circle minus moment of inertia of a circle about the same X axis.
- What is this moment of inertia of a circle, semicircle about this x ax?
- We know moment of inertia about its diameter axis is pi r to the power of 4 by 8.
- N R is capital R minus.
- Moment of initial circle, since as I said before, half of the circle is outside this semicircular area.
- Because this x axis is his diameter axis even for the smaller semicircle.
- That's why.
- R is radius of the smaller circle that is R by 4 to the power of 4 by 8.
- Then deduct that from this, we are going to get the value of IS.
- 3911 into r to the power of 4.
- This is the moment of inertia of the shaded area about the x axis, and this is when h is equal to 0.
- This is the answer when H is equals to 0.
- when h is equal to r by 2.
- This is the semicircle of radius r.
- This is the center of the semicircle and this is the axis of symmetry y axis.

Example or application from the lecture

- Let's get into the solution.
- Let's start with the first case that is when h is equal to 0.
- Let's start with the semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the center of the circle.

Key details

- about the x axis.
- Then how to find that?
- In that we have to detect moment of inertia of this circle about the same x axis.

- Then only we are going to get moment of inertia of this shaded area about the x axis.
- Moment of inertia of a circ semicircle, moment of inertia of a semicircle.
- about X axis minus.
- Moment of inertia I'll write here itself moment of inertia of a circle about the same x axis.
- I'll call it this one as I x1 and I'll call this as I x2.
- Then how to find moment of inertia of a semicircle about the x axis that is i x1 that will be equal to.
- Since the x axis itself is the diameter axis for the semicircle, moment of inertia we have a direct formula that is I r to the power of 4 by 8.
- Yet what is R?
- R is R itself.
- This is the moment of inertia of a semicircle above the.
- This is the moment of inertia of a circle about this line.
- Since we don't have a standard formula, we have to apply the parallel axis theorem.
- Where will be the centroid of this circle?
- Centroidal x-axis of this circle.
- This is the center of the circle.
- Draw a horizontal line passing through this.
- This will be the centroidal x-axis of a circle.
- We are talking about moment of inertia of this circle about this axis.
- Inertia about any line is equals to inertia about parallel centroidal axis plus area into distance square.
- into distance square.
- What is I bar x per circle?
- I bar x per circle is pi r to the power of 4 by 4 plus area.
- That is area of a circle.
- That is pi r square.
- D stands for the distance between the two axes.
- Here one axis is X and one more axis is X0.
- That is R by 2.
- Just substitute here pi into r is nothing but the radius of the circle.
- What is the radius of the circle?
- Look at the figure, it is r by 4.
- Plus pi into r square is r is radius of the circle.
- That is r by 4.
- Square into distance is r itself.
- This is r.
- We are going to get this as five.
- 17 pi r pi into 17 r to the power of 4 divided by 1024.
- This is the moment of inertia of the circle about the x axis.
- pi into r to the power of four divided by one zero to four.
- We are going to get this as 0.341 into R to the power of 4.
- This is the moment of inertia of the remaining.

Example or application from the lecture

- For that, we have to consider moment of inertia of this semicircle about x axis.
- Let's start with this solution.
- It is given in the figure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 15:00–19:59

area about the centroidal, sorry, about the x axis which is passing through the diameter axis.

Key details

- This is the value ix is equal to 0.341 into r to the power of 4.
- That's it for today's class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 41 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 41 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 41 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice